# Evolution Strategies with RL-style Gradient Estimation (ES-RL)

*A Mathematical Derivation and Literature Review*

Technical Document

**Abstract.** This document presents a self-contained mathematical exposition of the Evolution Strategy with Reinforcement-Learning-style gradient estimation (ES-RL) algorithm. Starting from the first principles of stochastic optimisation, we derive the score-function (REINFORCE) gradient estimator, motivate antithetic sampling as a variance-reduction technique, and derive the Adam adaptive moment update. The complete ES-RL algorithm is assembled in a unified framework; connections to natural evolution strategies and policy-gradient methods are established; and convergence properties are summarised. All results are anchored to primary literature references.

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# 1. Introduction and Motivation

Classical gradient-based optimisation requires access to $\nabla f(\mathbf{x})$, either analytically or via automatic differentiation. In many scientifically and industrially important settings this is not possible:

- The objective $f$ is a **black box** (a physics simulation, an experiment, a quantum circuit).
- $f$ is **non-differentiable** or **discontinuous**.
- The search space involves **discrete decisions** mixed with continuous parameters.
- The computational graph is too large for backpropagation.

Evolution Strategies (ES) address all of these cases by estimating the gradient of a *smoothed* version of $f$ using only function evaluations. The modern Reinforcement-Learning framing — due primarily to **Salimans et al. (2017)** [1] and **Wierstra et al. (2008, 2014)** [2, 3] — reinterprets ES as a policy-gradient algorithm on the distribution of candidate solutions, enabling the use of powerful optimisers such as **Adam** [4] developed for deep learning.

# 2. Problem Formulation

We seek to **maximise** a scalar objective:

$$\mathbf{x}^* = \arg\max_{\mathbf{x}\in\mathcal{X}} \; f(\mathbf{x}), \tag{1}$$

where $f : \mathcal{X} \subseteq \mathbb{R}^d \to \mathbb{R}$ may be **multimodal**, **noisy**, and **non-differentiable**. Direct access to $\nabla f$ is *not* assumed.

## 2.1 Smoothing via a Search Distribution

Rather than optimising $f$ directly, ES introduces a parametric family of probability distributions $\{p_{\boldsymbol{\theta}}\}_{\boldsymbol{\theta}}$ over $\mathcal{X}$ and maximises the **expected fitness**:

$$J(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{x}\sim p_{\boldsymbol{\theta}}}\Big[f(\mathbf{x})\Big] = \int_{\mathcal{X}} f(\mathbf{x})\, p_{\boldsymbol{\theta}}(\mathbf{x})\, d\mathbf{x}. \tag{2}$$

$J(\boldsymbol{\theta})$ is a **smoothed, everywhere differentiable** surrogate for $f$, even when $f$ itself is neither [5]. Maximising $J(\boldsymbol{\theta})$ concentrates probability mass near $\mathbf{x}^*$.

## 2.2 Isotropic Gaussian Search Distribution

Throughout this document we use the **isotropic Gaussian**:

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{N}\Big(\mathbf{x};\; \boldsymbol{\mu},\; \sigma^2\mathbf{I}\Big)\,, \tag{3}$$

where $\boldsymbol{\theta} = \boldsymbol{\mu} \in \mathbb{R}^d$ is the only optimisable parameter (the **mean** of the search distribution) and $\sigma > 0$ is a fixed **exploration spread**. The goal reduces to:

*Find the $\boldsymbol{\mu}$ that maximises $J(\boldsymbol{\mu})$.*

## 3. Evolution Strategies: Historical Background

Evolution Strategies were introduced independently by **Rechenberg** (1965, published 1973 [6]) and **Schwefel** (1965, published 1977 [7]) at the Technical University of Berlin. The original $(\mu, \lambda)$-ES:

1. Generates $\lambda$ offspring from $\mu$ parents by adding Gaussian noise.
2. Ranks offspring by fitness.
3. Selects the $\rho$ best as the next generation's parents.
4. Repeats.

Step-size self-adaptation was formalised by **Hansen and Ostermeier** in the Covariance Matrix Adaptation ES (CMA-ES) [8]. The modern **differentiable ES / Natural ES** framing [2, 3] reinterprets the fitness-weighted selection step as gradient ascent on $J(\boldsymbol{\theta})$, enabling any first-order optimiser.

## 4. The Score-Function Gradient Estimator

### 4.1 Derivation: The Log-Derivative Trick

To perform gradient ascent on $J(\boldsymbol{\mu})$ we need $\nabla_{\boldsymbol{\mu}} J$:

$$\nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu}) = \nabla_{\boldsymbol{\mu}} \int f(\mathbf{x})\, p_{\boldsymbol{\mu}}(\mathbf{x})\, d\mathbf{x}. \tag{4}$$

Under mild regularity conditions (Lebesgue dominated convergence) we may exchange the gradient and the integral:

$$= \int f(\mathbf{x})\, \nabla_{\boldsymbol{\mu}} p_{\boldsymbol{\mu}}(\mathbf{x})\, d\mathbf{x}. \tag{5}$$

Applying the **log-derivative identity**

$$\nabla_{\boldsymbol{\mu}} p_{\boldsymbol{\mu}}(\mathbf{x}) = p_{\boldsymbol{\mu}}(\mathbf{x})\, \nabla_{\boldsymbol{\mu}} \log p_{\boldsymbol{\mu}}(\mathbf{x}) \tag{6}$$

and substituting back, we obtain the **score-function estimator**:

**Key Result**

$$\nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu}) = \mathbb{E}_{\mathbf{x}\sim p_{\boldsymbol{\mu}}}\left[f(\mathbf{x}) \cdot \nabla_{\boldsymbol{\mu}} \log p_{\boldsymbol{\mu}}(\mathbf{x})\right]. \tag{7}$$

This is also known as the **REINFORCE gradient** in reinforcement learning [9] and the **natural gradient direction** in natural evolution strategies [2, 3].

**Remark 4.1.** *The right-hand side of* (7) *is an expectation estimable by Monte Carlo sampling — requiring* only function evaluations*, not analytic gradients of $f$.*

### 4.2 Closed Form for the Isotropic Gaussian

For $p_{\boldsymbol{\mu}}(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \sigma^2\mathbf{I})$:

$$\log p_{\boldsymbol{\mu}}(\mathbf{x}) = -\tfrac{d}{2}\log(2\pi\sigma^2) - \frac{\|\mathbf{x}-\boldsymbol{\mu}\|^2}{2\sigma^2}. \tag{8}$$

$$\nabla_{\boldsymbol{\mu}} \log p_{\boldsymbol{\mu}}(\mathbf{x}) = \frac{\mathbf{x}-\boldsymbol{\mu}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}}{\sigma}, \qquad \text{where } \mathbf{x} = \boldsymbol{\mu} + \sigma\boldsymbol{\varepsilon},\ \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{9}$$

Substituting into (7):

$$\nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu}) = \frac{1}{\sigma}\,\mathbb{E}_{\boldsymbol{\varepsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon})\cdot\boldsymbol{\varepsilon}]\,. \tag{10}$$

### 4.3 Monte Carlo Estimator

Using $H$ independent samples $\{\boldsymbol{\varepsilon}_i\}_{i=1}^H$:

$$\hat{\mathbf{g}}_H = \frac{1}{H\sigma}\sum_{i=1}^{H} f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon}_i)\,\boldsymbol{\varepsilon}_i. \tag{11}$$

**Proposition 4.1** (Unbiasedness)**.** $\mathbb{E}[\hat{\mathbf{g}}_H] = \nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu})$.

The variance scales as:

$$\mathrm{Var}[\hat{\mathbf{g}}_H] = \frac{1}{H\sigma^2}\mathrm{Var}_{\boldsymbol{\varepsilon}}[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon})\,\boldsymbol{\varepsilon}] = O\Big(\tfrac{1}{H}\Big)\,, \tag{12}$$

motivating the use of as many perturbations as the budget allows.

## 5. Antithetic Sampling: Variance Reduction

### 5.1 Motivation

The estimator (11) has variance $O(1/H)$. For finite $H$, individual estimates can be very noisy when $f$ is multimodal or when $\sigma$ is small relative to the scale of the landscape.

### 5.2 Antithetic Pairs

The **antithetic variates** method [10] exploits the symmetry of the Gaussian by evaluating **symmetric pairs** $(\boldsymbol{\varepsilon}_i, -\boldsymbol{\varepsilon}_i)$:

$$\hat{\mathbf{g}}_{\text{anti}} = \frac{1}{2H\sigma}\sum_{i=1}^{H}\Big[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon}_i) - f(\boldsymbol{\mu}-\sigma\boldsymbol{\varepsilon}_i)\Big]\,\boldsymbol{\varepsilon}_i. \tag{13}$$

### 5.3 Proof of Unbiasedness

*Proof.* Since $\boldsymbol{\varepsilon}_i \overset{d}{=} -\boldsymbol{\varepsilon}_i$ (Gaussian symmetry):

$$\begin{aligned}
\mathbb{E}[\hat{\mathbf{g}}_{\text{anti}}] &= \frac{1}{2\sigma}\mathbb{E}[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}] + \frac{1}{2\sigma}\mathbb{E}[-f(\boldsymbol{\mu}-\sigma\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}] \\
&= \frac{1}{2\sigma}\mathbb{E}[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}] + \frac{1}{2\sigma}\mathbb{E}[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}] \\
&= \frac{1}{\sigma}\mathbb{E}[f(\boldsymbol{\mu}+\sigma\boldsymbol{\varepsilon})\boldsymbol{\varepsilon}] = \nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu}). \qquad \checkmark
\end{aligned} \tag{14}$$

The second equality uses the substitution $\boldsymbol{\varepsilon} \mapsto -\boldsymbol{\varepsilon}$, which leaves the Gaussian distribution invariant. □

### 5.4 Variance Reduction Analysis

Let $A_i = f(\boldsymbol{\mu} + \sigma\boldsymbol{\varepsilon}_i)\varepsilon_i$ and $B_i = -f(\boldsymbol{\mu} - \sigma\boldsymbol{\varepsilon}_i)\varepsilon_i$. The antithetic estimate uses $(A_i + B_i)/2$ instead of $A_i$. For smooth $f$, a positive perturbation that yields a high value makes the mirror perturbation tend to yield a lower value; hence $\mathrm{Cov}(A_i, B_i) < 0$, and:

$$\mathrm{Var}\left[\frac{A_i + B_i}{2}\right] = \frac{\mathrm{Var}[A_i] + \mathrm{Var}[B_i] + 2\mathrm{Cov}(A_i, B_i)}{4} < \frac{\mathrm{Var}[A_i]}{2}. \tag{15}$$

**Key Result**

Antithetic sampling **at least halves the estimator variance** relative to one-sided sampling, at no extra cost in the number of perturbation *directions* $H$ (though $2H$ total evaluations are required).

This was demonstrated empirically by Salimans et al. [1], who showed that antithetic sampling roughly doubles effective sample efficiency in high-dimensional problems.

## 6. The Adam Optimiser

### 6.1 Motivation

Applying the raw estimate $\hat{\mathbf{g}}$ in a fixed-step update $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + \alpha\hat{\mathbf{g}}$ suffers from:

- **Sensitivity to $\alpha$:** too large $\Rightarrow$ oscillation; too small $\Rightarrow$ slow convergence.
- **Non-stationarity:** the optimal step size changes as $\boldsymbol{\mu}$ approaches the optimum.
- **Noise amplification:** estimates from a small batch of perturbations are inherently noisy.

**Adam** (Adaptive Moment Estimation) [4] addresses all three by maintaining per-parameter running statistics of the gradient.

### 6.2 Derivation from Exponential Moving Averages

Let $\hat{\mathbf{g}}_t$ be the gradient estimate at step $t$. Define:

**First moment** (momentum):

$$\mathbf{m}_t = \beta_1\,\mathbf{m}_{t-1} + (1 - \beta_1)\,\hat{\mathbf{g}}_t, \qquad \mathbf{m}_0 = \mathbf{0}. \tag{16}$$

**Second moment** (uncentred variance):

$$\mathbf{v}_t = \beta_2\,\mathbf{v}_{t-1} + (1 - \beta_2)\,\hat{\mathbf{g}}_t^2, \qquad \mathbf{v}_0 = \mathbf{0}, \tag{17}$$

where $\beta_1, \beta_2 \in [0, 1)$ and squaring is elementwise.

### 6.3 Bias Correction

Unrolling (16) from $\mathbf{m}_0 = \mathbf{0}$:

$$\mathbf{m}_t = (1-\beta_1)\sum_{i=1}^{t}\beta_1^{t-i}\,\hat{\mathbf{g}}_i. \tag{18}$$

Taking expectations under the stationarity assumption $\mathbb{E}[\hat{\mathbf{g}}_i] = \mathbf{g}$:

$$\mathbb{E}[\mathbf{m}_t] = \mathbf{g}\,(1-\beta_1)\sum_{i=1}^{t}\beta_1^{t-i} = \mathbf{g}\,(1-\beta_1^t). \tag{19}$$

The estimate is **biased toward zero** by the factor $(1-\beta_1^t)$. Analogously, $\mathbb{E}[\mathbf{v}_t] = \mathbb{E}[\mathbf{g}^2]\,(1-\beta_2^t)$. We correct with:

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1-\beta_1^t}, \qquad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1-\beta_2^t}, \tag{20}$$

so that $\mathbb{E}[\hat{\mathbf{m}}_t] \approx \mathbf{g}$ and $\mathbb{E}[\hat{\mathbf{v}}_t] \approx \mathbb{E}[\mathbf{g}^2]$.

### 6.4 The Update Rule

**Key Result**

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t + \alpha \cdot \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \varepsilon}, \tag{21}$$

where $\varepsilon \approx 10^{-8}$ prevents division by zero.

Table 1: Interpretation of the Adam update terms.

| Term | Role |
|---|---|
| $\hat{\mathbf{m}}_t$ | Smoothed gradient direction; reduces noise via momentum, suppresses short-lived fluctuations. |
| $\sqrt{\hat{\mathbf{v}}_t}$ | Running RMS of gradient magnitude; normalises the step size per coordinate. |
| $\alpha/\sqrt{\hat{\mathbf{v}}_t}$ | Per-parameter adaptive rate: smaller for high-gradient coordinates, larger for low-gradient or infrequently-updated ones. |

### 6.5 Standard Hyperparameters

## 7. The Complete ES-RL Algorithm

### 7.1 State Variables

### 7.2 One Iteration

**Step 1 — Sample perturbations:**

$$\varepsilon_1, \ldots, \varepsilon_H \overset{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d).$$

Table 2: Recommended Adam hyperparameters [4].

| Parameter | Typical value | Role |
|---|---|---|
| $\alpha$ | $10^{-3}$–0.05 | Global learning rate (primary tuning knob) |
| $\beta_1$ | 0.9 | Momentum decay |
| $\beta_2$ | 0.999 | RMS decay |
| $\varepsilon$ | $10^{-8}$ | Numerical stability |

Table 3: ES-RL state variables.

| Variable | Dimension | Description |
|---|---|---|
| $\boldsymbol{\mu}$ | $d$ | Current mean of the search distribution |
| $\sigma$ | scalar | Exploration spread (fixed) |
| $\mathbf{m}$ | $d$ | Adam first-moment accumulator |
| $\mathbf{v}$ | $d$ | Adam second-moment accumulator |
| $t$ | integer | Adam timestep counter |

**Step 2 — Evaluate antithetic pairs** ($2H$ evaluations):

$$f_i^+ = f(\boldsymbol{\mu} + \sigma\boldsymbol{\varepsilon}_i), \quad f_i^- = f(\boldsymbol{\mu} - \sigma\boldsymbol{\varepsilon}_i), \qquad i = 1, \ldots, H.$$

**Step 3 — Gradient estimate** (eq. (13)):

$$\hat{\mathbf{g}} = \frac{1}{2H\sigma} \sum_{i=1}^{H} (f_i^+ - f_i^-)\, \boldsymbol{\varepsilon}_i.$$

**Step 4 — Adam moment update:**

$$t \leftarrow t + 1, \quad \mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1)\hat{\mathbf{g}}, \quad \mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1 - \beta_2)\hat{\mathbf{g}}^2.$$

**Step 5 — Bias correction:**

$$\hat{\mathbf{m}} = \mathbf{m}/(1 - \beta_1^t), \qquad \hat{\mathbf{v}} = \mathbf{v}/(1 - \beta_2^t).$$

**Step 6 — Update mean:**

$$\boldsymbol{\mu} \leftarrow \mathrm{clip}\left(\boldsymbol{\mu} + \alpha \cdot \frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + \varepsilon},\ \mathbf{x}_{\min},\ \mathbf{x}_{\max}\right).$$

### 7.3 Full Pseudocode

**Algorithm 1** ES-RL with Antithetic Sampling and Adam

**Require:** $f$, $\mathcal{X}=[\mathbf{x}_{\min}, \mathbf{x}_{\max}]$, $\sigma$, $H$, $\alpha$, $\beta_1$=0.9, $\beta_2$=0.999, $\varepsilon=10^{-8}$, budget $N$

$\boldsymbol{\mu} \leftarrow$ random point in $\mathcal{X}$
$\mathbf{m} \leftarrow \mathbf{0}$; $\mathbf{v} \leftarrow \mathbf{0}$; $t \leftarrow 0$ ▷ Adam state
$f^* \leftarrow -\infty$; $\mathbf{x}^* \leftarrow$ None
**while** evaluations $< N$ **do**
  Sample $\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_H \overset{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I})$
  **for** $i = 1, \ldots, H$ **do**
    $f_i^+ \leftarrow f(\text{clip}(\boldsymbol{\mu} + \sigma\boldsymbol{\varepsilon}_i, \mathbf{x}_{\min}, \mathbf{x}_{\max}))$
    $f_i^- \leftarrow f(\text{clip}(\boldsymbol{\mu} - \sigma\boldsymbol{\varepsilon}_i, \mathbf{x}_{\min}, \mathbf{x}_{\max}))$
    Update $f^*, \mathbf{x}^*$ if $\max(f_i^+, f_i^-) > f^*$
  **end for**
  $\hat{\mathbf{g}} \leftarrow \dfrac{1}{2H\sigma} \sum_{i=1}^{H} (f_i^+ - f_i^-)\, \boldsymbol{\varepsilon}_i$
  $t \leftarrow t + 1$
  $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1-\beta_1)\hat{\mathbf{g}}$; $\mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1-\beta_2)\hat{\mathbf{g}}^2$
  $\hat{\mathbf{m}} \leftarrow \mathbf{m}/(1-\beta_1^t)$; $\hat{\mathbf{v}} \leftarrow \mathbf{v}/(1-\beta_2^t)$
  $\boldsymbol{\mu} \leftarrow \text{clip}\Big(\boldsymbol{\mu} + \alpha\, \hat{\mathbf{m}}/(\sqrt{\hat{\mathbf{v}}} + \varepsilon),\ \mathbf{x}_{\min}, \mathbf{x}_{\max}\Big)$
**end while**
**return** $\mathbf{x}^*, f^*$

## 8. Convergence Analysis

### 8.1 Relationship to Gaussian Smoothing

The expected fitness $J(\boldsymbol{\mu})$ equals the Gaussian convolution of $f$:

$$J(\boldsymbol{\mu}) = (f * \phi_\sigma)(\boldsymbol{\mu}), \qquad \phi_\sigma(\mathbf{x}) = \mathcal{N}(\mathbf{x};\, \mathbf{0}, \sigma^2\mathbf{I}). \tag{22}$$

Nesterov and Spokoiny [5] showed that $J$ inherits Lipschitz smoothness from $f$: if $f$ is $L$-Lipschitz, then $J$ is $L$-Lipschitz and its gradient is $(L/\sigma)$-Lipschitz, guaranteeing convergence of gradient ascent on $J$.

### 8.2 Strongly Concave Case

For $J$ that is $\kappa$-strongly concave near the maximum, projected gradient ascent with step $\alpha$ converges geometrically [11]:

$$J(\boldsymbol{\mu}^*) - J(\boldsymbol{\mu}_t) \;\le\; (1-\alpha\kappa)^t \Big[J(\boldsymbol{\mu}^*) - J(\boldsymbol{\mu}_0)\Big]. \tag{23}$$

In practice $J$ inherits the multimodality of $f$ and convergence to a *local* maximum is what is guaranteed.

### 8.3 Finite-Sample Rate (Non-Convex)

With $H$ antithetic pairs, the gradient estimate has variance $O(1/H)$. By the non-convex stochastic gradient theory [12]:

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\Big[\big\|\nabla J(\boldsymbol{\mu}_t)\big\|^2\Big] = O\left(\frac{1}{\sqrt{T}} + \frac{\sigma_g}{\sqrt{H}}\right), \tag{24}$$

where $\sigma_g$ is the variance of one gradient sample. Larger $H$ tightens the bound.

### 8.4 Approximation Bias from $\sigma$

A Taylor expansion [1] shows:

$$\nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu}) = \nabla f(\boldsymbol{\mu}) + O(\sigma^2). \tag{25}$$

Small $\sigma$: nearly unbiased, but high variance and inability to escape local maxima. Large $\sigma$: low variance but the smoothed surrogate may miss narrow peaks. This tension is the fundamental exploration–exploitation trade-off in ES.

## 9. Connections to Related Methods

### 9.1 REINFORCE (Williams, 1992)

Williams [9] derived the score-function estimator in reinforcement learning, where $f(\mathbf{x})$ is a **return** and $p_{\boldsymbol{\theta}}(\mathbf{x})$ is a **stochastic policy**:

$$\hat{\mathbf{g}}_{\text{REINFORCE}} = \frac{1}{H} \sum_{i=1}^{H} R(\tau_i)\, \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\tau_i). \tag{26}$$

Substituting $R \equiv f$, $\pi_{\boldsymbol{\theta}} \equiv p_{\boldsymbol{\mu}}$, $\tau_i \equiv \mathbf{x}_i$ recovers eq. (11) exactly. ES-RL is a direct instantiation of REINFORCE with a Gaussian "policy."

### 9.2 Natural Evolution Strategies (NES)

Wierstra et al. [2, 3] proposed the **natural gradient** update:

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t + \alpha\, \mathbf{F}^{-1}(\boldsymbol{\mu}_t)\, \hat{\mathbf{g}}, \tag{27}$$

where $\mathbf{F}(\boldsymbol{\mu}) = \mathbb{E}\Big[(\nabla_{\boldsymbol{\mu}} \log p_{\boldsymbol{\mu}})(\nabla_{\boldsymbol{\mu}} \log p_{\boldsymbol{\mu}})^\top\Big]$ is the Fisher information matrix. For the isotropic Gaussian $\mathbf{F} = \sigma^{-2}\mathbf{I}$, which simplifies to $\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t + (\alpha/\sigma)\hat{\mathbf{g}}_{\text{norm}}$, removing the scale dependence on $\sigma$.

### 9.3 CMA-ES

CMA-ES [8] adapts a full covariance matrix $\mathbf{C}$ giving second-order-like convergence at $O(d^2)$ cost per step. ES-RL with an isotropic Gaussian is a first-order $O(d)$ variant, trading convergence speed for scalability.

## 10. Summary and Practical Guidance

### 10.1 Algorithm in One Equation

**ES-RL Update (complete)**

$$\boldsymbol{\mu}_{t+1} = \text{clip}\left(\boldsymbol{\mu}_t + \alpha \cdot \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \varepsilon},\ \mathbf{x}_{\min},\ \mathbf{x}_{\max}\right), \tag{28}$$

where $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are the bias-corrected Adam moments of the antithetic gradient estimate (13).

### 10.2 Hyperparameter Guidance

Table 4: ES-RL hyperparameter guidance.

| Hyperparameter | Recommended | Effect of increasing |
|---|---|---|
| $\sigma$ (spread) | 0.05–0.20 (fraction of domain) | Better exploration; more approximation bias |
| $H$ (pairs/step) | 5–20 | Lower gradient variance; more evaluations per step |
| $\alpha$ (learning rate) | 0.01–0.10 | Faster but potentially less stable convergence |
| $\beta_1$ (momentum) | 0.9 (fixed) | Smoother gradient direction |
| $\beta_2$ (RMS decay) | 0.999 (fixed) | More stable norm estimation |

### 10.3 When ES-RL Outperforms Random Search

**ES-RL is advantageous when:**

- The landscape has a **clear basin** around the optimum (gradient signal is informative).
- The **evaluation budget is limited** relative to domain size.
- Dimensionality $d > 1$ (random search needs $O((1/p_\varepsilon)^{1/d})$ samples — exponential in $d$).

**Random search remains competitive when:**

- The landscape is **pathological** (needle-in-haystack; gradient information is useless).
- The domain is small or **1-dimensional** with modest budget.
- **Global diversity** is more important than local convergence.

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